
Session 4 — Linear Algebra II

What today is about — the payoff session

Today the two halves of the course meet.

- *Eigenvectors* and *eigenvalues*: the special directions of a transformation.
- **Principal Component Analysis**: the flagship tool for reducing and visualising high-dimensional data.
- The *SVD*: the general engine behind PCA.
- *Dynamics*: how eigenvalues decide whether a system settles, oscillates, or explodes.

Key idea

The **covariance matrix** from Session 2 (a statistical object) read through its **eigenvectors** (a linear-algebra tool) *is* Principal Component Analysis. That single sentence is the destination of today.

So keep the covariance matrix in mind as we build the eigenvalue machinery — we are assembling exactly the pieces PCA needs.

The special directions of a matrix

Most vectors get knocked off their direction when a matrix acts on them. A few special ones do not.

Definition

A nonzero vector v is an **eigenvector** of A if A only *stretches* it, without changing its direction:

$$Av = \lambda v,$$

where the number λ is the **eigenvalue** — the stretch factor.

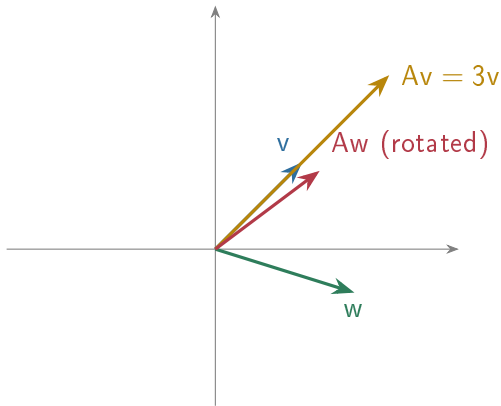
Eigenvectors are the “grain of the wood”

Intuition

Think of the transformation as flowing space around. Almost every arrow gets rotated as well as stretched. But along an eigenvector, the transformation is pure stretch: the arrow stays on its own line, only getting longer, shorter, or flipped.

Along these special directions a complicated matrix behaves like plain multiplication by a number. That simplicity is what makes eigenvectors so powerful.

Eigenvectors — the picture



Along the eigenvector v the matrix only stretches (here by 3). A generic vector w gets rotated off its line — it is not an eigenvector.

How to find them

To find eigenvalues we solve the **characteristic equation** $\det(A - \lambda I) = 0$. Each root λ then gives its eigenvectors by solving $(A - \lambda I)v = 0$.

Example

For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$:

$$\det \begin{pmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{pmatrix} = (2 - \lambda)^2 - 1 = 0 \Rightarrow \lambda = 3 \text{ or } 1.$$

The eigenvectors are $(1, 1)$ for $\lambda = 3$ and $(1, -1)$ for $\lambda = 1$ — and they are *orthogonal*.

Two free checks on your eigenvalues

Useful fact

For any square matrix:

$$\sum_i \lambda_i = \text{tr}(A) \text{ (sum of the diagonal),} \quad \prod_i \lambda_i = \det(A).$$

For the example above: $\lambda_1 + \lambda_2 = 3 + 1 = 4 = \text{tr}(A)$, and $\lambda_1 \lambda_2 = 3 \cdot 1 = 3 = \det(A)$.
Always use these to catch arithmetic slips before going any further.

Why eigenvalues make repeated action easy

Diagonalization

Collect the eigenvectors as columns of P and the eigenvalues on the diagonal of D ; then $A = PDP^{-1}$, and powers become trivial:

$$A^k = PD^kP^{-1}, \quad D^k = \text{diag}(\lambda_1^k, \dots, \lambda_n^k).$$

In the eigenvector coordinates the matrix is just a set of independent stretches. This is why the long-run behaviour of A^k is governed entirely by the eigenvalues — exactly what we need for dynamics later today.

Symmetric matrices are especially nice

Key idea

If $A = A^T$ (symmetric — as every covariance matrix is), then its eigenvalues are real and its eigenvectors are mutually orthogonal. The transformation is then just a set of independent stretches along perpendicular axes.

A symmetric matrix is *positive-definite* when all its eigenvalues are positive. Covariance matrices are positive-semidefinite — their eigenvalues are never negative, because they represent amounts of variance. This clean structure is exactly what makes PCA work.

Practice — try it

Let $A = \begin{pmatrix} 3 & 0 \\ 0 & -2 \end{pmatrix}$ (already diagonal).

1. What are its eigenvalues and eigenvectors?
2. Check the trace and the determinant against your eigenvalues.
3. What does A do to the vector $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$?

The problem PCA solves

High-dimensional data are hard to see and often redundant. Because many measured variables are correlated, the cloud of data points really lies close to a lower-dimensional subspace.

PCA finds the few directions that capture the most variation, so we can describe, visualise, and compress the data with little loss. It answers: *along which directions does the data actually vary?*

Key idea

The principal components are the **eigenvectors of the covariance matrix**, and each eigenvalue is the **amount of variance** along that direction. The first component is the direction of greatest spread; the next is the greatest remaining spread orthogonal to it; and so on.

Everything from the first half of today — eigenvectors, symmetric matrices, orthogonality — comes together in this one idea.

The PCA recipe

1. **Centre** the data: subtract each variable's mean, so the cloud sits at the origin.
2. Form the **covariance matrix** C of the centred data.
3. Compute the **eigenvectors and eigenvalues** of C .
4. Order the eigenvectors by eigenvalue, largest first: these are principal components $1, 2, 3, \dots$
5. **Project** the data onto the top few components to reduce dimension.

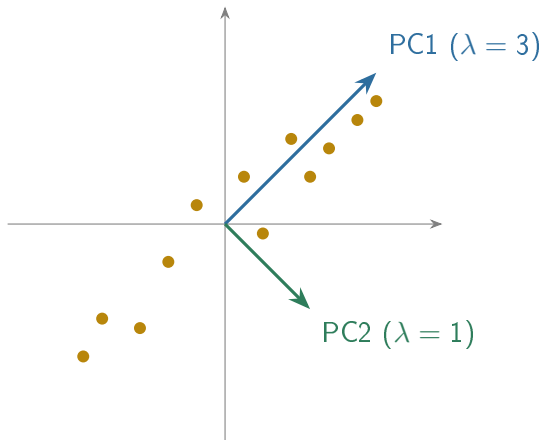
What PCA is doing, geometrically

Intuition

Picture an elongated, tilted cloud of points. PCA finds the long axis of the cloud (the direction of most variance — the first component) and the short axis perpendicular to it (the second).

Rotating onto these axes “straightens” the cloud so its directions become uncorrelated. Keeping only the long axis is dimensionality reduction: we describe each point by its position along the direction that matters most.

PCA — the picture



PC1 follows the long axis of the cloud (most variance); PC2 is perpendicular, capturing the little that remains.

Example

Two mean-centred variables have covariance matrix $C = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

We already found its eigenpairs: $\lambda_1 = 3$ along $(1, 1)$, and $\lambda_2 = 1$ along $(1, -1)$. So the first principal component points along $(1, 1)$ (the two variables rise together) and carries variance 3; the second carries variance 1. The fraction of variance explained by the first component is

$$\frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{3}{4} = 75\%.$$

Useful fact

The **total variance** is the sum of the eigenvalues, equal to the trace of C . The **proportion explained** by component j is $\lambda_j / \sum_i \lambda_i$; plotting these in decreasing order gives the **scree plot**, whose “elbow” suggests how many components to keep. Because C is a covariance matrix, all $\lambda_i \geq 0$.

A caution about PCA

Common pitfall

PCA is a *linear* method and is sensitive to *scaling*. A variable measured in large units (say reaction time in milliseconds) can dominate the components purely because of its units. Standardising the variables, or using the correlation matrix, is often necessary.

Also remember: PCA finds directions of *variance*, which are not always the directions that matter for a task. Variance is not the same as relevance.

Link to cognitive science

PCA reduces recordings of hundreds of neurons to a few “latent” dimensions along which population activity evolves (neural state-space analysis); it compresses fMRI voxels and EEG channels; it extracts the principal components of face images (“eigenfaces”); and it summarises questionnaire batteries into a few underlying factors.

Whenever someone says a system is “effectively low-dimensional”, PCA is usually how they found out.

Practice — try it

After PCA on some neural data, the covariance has eigenvalues $\lambda = (6, 3, 1)$ along three orthogonal components.

1. What is the total variance?
2. What fraction does the first component explain?
3. How many components are needed to keep at least 85% of the variance?

The SVD: the general engine behind PCA

Definition

The **Singular Value Decomposition** writes *any* matrix as $X = U\Sigma V^T$: a rotation, a set of non-negative stretches (the *singular values* in Σ), and another rotation.

The idea to carry away: *every* linear transformation is a rotation, then a scaling along perpendicular axes, then another rotation. PCA is just the SVD applied to the centred data matrix, and the singular values squared are the variances. SVD also handles non-square data, which PCA's covariance route cannot.

Eigenvalues also predict the future

Many models describe how a state x evolves, either step by step or continuously in time. Eigenvalues decide what happens.

Key idea

Discrete time $x_{t+1} = Ax_t$: stable (shrinks to zero) if every $|\lambda| < 1$; blows up if some $|\lambda| > 1$.

Continuous time $\dot{x} = Ax$: stable if every eigenvalue has *negative real part*; imaginary parts produce oscillations.

Why eigenvalues govern stability

Intuition

Eigenvalues are the growth rates of the natural “modes” of a system. Decompose any starting state into its eigenvector components; each then evolves independently, growing or decaying at its own rate. The largest eigenvalue eventually dominates and sets the long-run behaviour.

This is why the eigenvalues alone tell you whether a dynamical system settles, oscillates, or explodes — without solving anything in detail.

Example

A discrete system $x_{t+1} = Ax_t$ has eigenvalues $\lambda_1 = 0.5$ and $\lambda_2 = 1.2$. What happens over time?

Along the first eigenvector the state is multiplied by 0.5 each step, so that component decays to zero. Along the second it is multiplied by 1.2, growing by 20% each step. Because one $|\lambda| > 1$, the system is **unstable**: almost every starting state eventually grows without bound. It would be stable only if *both* eigenvalues had magnitude below 1.

Link to cognitive science

Recurrent neural networks and models of neural population dynamics are exactly systems of the form $\dot{x} = Ax$ (plus inputs and nonlinearities). Whether a memory persists, an activity pattern decays, or a network oscillates is governed by the eigenvalues of its connectivity matrix.

This also closes the loop with the differential-equations material elsewhere in the programme: the qualitative behaviour of a linear system is read straight off its eigenvalues.

Practice — try it

Stable or unstable?

1. A continuous system with eigenvalues -1 and -3 .
2. A continuous system with eigenvalues -2 and $+0.5$.
3. A discrete system with eigenvalues 0.9 and 0.4 .

Session 4 — what we built

- *Eigenvectors* are the directions a matrix only stretches, by factor λ ; symmetric matrices have real eigenvalues and orthogonal eigenvectors.
- *PCA* uses the eigenvectors of the covariance matrix as principal components, and the eigenvalues as the variances along them.
- The *SVD* is the general rotation–scale–rotation behind PCA.
- In *dynamics*, eigenvalues set the growth or decay of each mode, and thus stability.

Next: probability again — now armed with vectors and matrices.